ARGO-GIM

AUTONOMOUS REAL-TIME GRADIENT OVERLAY · GASTRO INTESTINAL MONITOR

CONCEPT & R&D · TALENTIS 2026 BEST RUNNER-UP

Your gut, seen from the inside. At home, with no invasive preparation.

An ingestible capsule concept combining high-resolution optical cameras, biochemical biosensor arrays, and on-board Edge-AI for longitudinal gut health monitoring and colorectal cancer early prevention.

CLINICAL & EPIDEMIOLOGICAL PROBLEM

Current screening fails to reach those who need it most

Colorectal cancer is the second leading cause of cancer death in Europe. When diagnosed at a localized stage, 5-year survival reaches 91%, but plummets to 13% once metastases occur. In Italy, screening adherence averages only 35.8% (dropping to 21.1% in the South and Islands) due to fear of procedural invasiveness and demanding bowel preparation regimes.

35.8%
Italian screening uptake (ONS 2024)

21.1%
Uptake in Southern Italy & Islands

91% / 13%
5-year survival: localized vs metastatic

THE ARGO-GIM SOLUTION

Integrated 4-component system architecture

ARGO-GIM (Autonomous Real-time Gradient Overlay - Gastro Intestinal Monitor) is designed to transform an invasive hospital procedure into a non-invasive, home-based examination:

- **Miniaturized capsule (28 × 12 mm):** dual-end cameras, biosensor array, ultra-low-power radio, and solid-state biocompatible battery.
- **Lightweight sensor vest:** worn comfortably under daily clothes to track telemetry packets and anatomically localize the capsule.
- **Compact receiver unit:** records physiological signals during normal daily activities.
- **Companion App & AI dashboard:** computes a gut health index and compiles a structured clinical report.

TECHNOLOGICAL INNOVATION · DIFFERENTIATION FROM CONVENTIONAL CAPSULES (E.G. PILLCAM™)

FEATURE	CONVENTIONAL ENDOSCOPY CAPSULES (E.G. PILLCAM™)	ARGO-GIM INNOVATION & ADVANTAGES
Bowel Preparation	`Mandatory & Demanding` Intensive laxatives and fasting required to clear optics.	`Non-Invasive` **Biosensors operate even within physiological contents, removing adoption barriers.**
Data Modality	`Optical video only` Surface photography; blind to biochemical shifts or sub-visual inflammation.	`Multimodal` **High-res optics + biochemical biosensors (pH, oxidative stress ROS, haemoglobin, temp).**
Frame Processing	`Up to 50,000 raw frames` Hours of manual physician frame-by-frame review.	`On-board Edge-AI` **Local filtering of repetitive frames: transmits only suspicious anomalies, saving time and power.**
Setting of Use	Hospital specialist outpatient procedure.	**Home-based longitudinal monitoring (Lite) and advanced hospital diagnostics (Dx).**

ARGO-GIM LITE · HOME PREVENTION

Tailored for proactive individuals tracking gut health over time. On-board AI identifies baseline shifts, alerting users to seek timely medical consultation before macroscopic lesions develop.

ARGO-GIM DX · CLINICAL SPECIALIST DEVICE

Equipped with Narrow-Band Imaging (NBI) and structured-light 3D depth mapping. AI-assisted lesion classification (Paris, Kudo, NICE) and native FHIR hospital EHR interoperability.

ARGO-GIM is an early-stage technological and scientific concept, not an approved medical device. Specifications represent design targets to be clinically validated.
Sources: [1] Italian National Screening Observatory (ONS 2024) · [2] American Cancer Society, SEER data · © 2026 ARGO-GIM · info@argogim.com

ARGO-GIM

EXECUTIVE SUMMARY · 2/2 · DEVELOPMENT ROADMAP & REQUIREMENTS

TRACK RECORD · RECOGNITIONS & ACCELERATION PROGRAMS

01 · ACCELERATION & LAB

CAL.HUB.RIA - LAB

Talent, ideas, value for the future: biomedical incubation and scientific validation program.

02 · INNOVATION CONTEST

Talentis · Maratea

Competitive selection and pitch presentation at the national innovation competition.

03 · NATIONAL FINALS

Talentis · CAPRI 2026

Awarded as **Best Runner-up**, securing qualification for the national finals in Capri.

CONCRETE TARGETS · BENCHTOP PROTOTYPE (12 MONTHS)

Our 12-month operational plan targets achieving **TRL 4 (functional benchtop prototype validated in lab)**:

- A **Circuit Integration:** rigid-flex PCB layout integrating ultra-low-power Cortex-M MCU and RF transceiver.
- B **Biosensor Interface:** functional micro-electrodes for pH, haemoglobin, and oxidative stress (ROS) detection.
- C **In-vitro Gut Simulator:** fluidic testing in anatomical phantoms to evaluate buoyancy, optics, and RF transmission through tissue.
- D **Intellectual Property:** patent filing covering the integrated multimodal sensing architecture and Edge-AI filtering pipeline.

WHAT WE SEEK · TECHNICAL PARTNERSHIPS

To advance the prototype development, we actively seek collaborations with key technical profiles and research partners:

- 01 **TECHNICAL CO-FOUNDER:** Electronic & Biomedical Engineering / Embedded Firmware to lead hardware integration and sensor design.
- 02 **Gastroenterology Units:** clinical partners to co-develop validation protocols and participate in pilot studies.
- 03 **MedTech Regulatory Mentors:** guidance for EU MDR 2017/745 pathway and ISO 10993 biocompatibility standards.
- 04 **Microelectronic Laboratories:** research partners offering testing benches and anechoic chambers.

REGULATORY PATHWAY & DEVELOPMENT PHASES (EU MDR 2017/745)

PHASE 1 · TODAY
Concept & Design
3D CAD, simulations, and sensor selection.

PHASE 2
TRL 4 Prototype
Hardware integration, in-vitro testing, and patent.

PHASE 3
Preclinical
ISO 10993 biocompatibility & electrical safety.

PHASE 4
Clinical Trials
Feasibility and comparative clinical validation.

PHASE 5
CE Marking
Medical device clearance for Lite & Dx.

THE ARGO-GIM DEVELOPMENT TEAM

NV **Niccolò Vono** · CO-FOUNDER
PhD Student in Molecular Medicine Biotechnology · colorectal cancer multi-omics

CV **Cristina Vocca** · CO-FOUNDER
PhD Student in Public Health · resident in Clinical Pharmacology and Toxicology

GF **Giuseppina Fruci** · CO-FOUNDER
Master's Student in Pharmacy

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